INCH-POUND

MIL-R-83579/1
6 April 1994

ASSOCIATED DETAIL SPECIFICATION

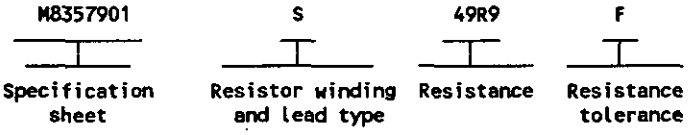
RESISTORS, FIXED, WIRE-WOUND (POWER TYPE),
HIGH RELIABILITY,
STYLE RWS81

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RWS81, high reliability, power type, wire-wound, fixed resistor. These resistors are available with terminal types "S" and "W" (solderable or weldable, inductively wound) and terminal "N" (solderable, noninductively wound), and terminal "Z" (weldable, noninductively wound).

1.2 Part or Identifying Number (PIN). The PIN shall be in the following form, and as specified in MIL-R-83579.



2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-83579 - Resistors, Fixed, Wire-Wound (Power Type), High Reliability, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, Robins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SMC/SDFP, P.O. Box 92960, Worldway Postal Center, Los Angeles, CA 90009-2960, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

AMSC N/A FSC 5905
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3. REQUIREMENTS

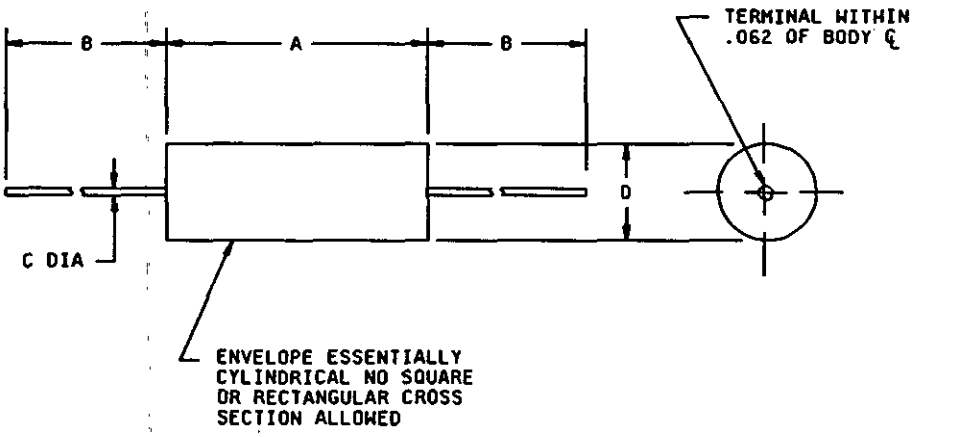
3.1 Requirements. Requirements shall be in accordance with MIL-R-83579, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Noninductive windings. Noninductively wound resistors are identified by terminal designation "N" to indicate a solderable terminal, and "Z" to indicate a weldable terminal, noninductively wound resistor.

3.3 Minimum resistance value for "S" and "W" (inductively) wound type resistors. For inductively wound and noninductively wound resistors, the minimum resistance value shall be 0.1 ohm.

3.4 Maximum resistance value for "S" and "W" type resistors. The maximum resistance value based on use of .001 inch (0.03 mm) nominal diameter wire (with a nominal resistivity of 800 ohms per circular mil foot) shall be 492 ohms. The maximum resistance value based on use of .0008 inch (0.020 mm) nominal diameter wire (.00072 inch (0.0183 mm) absolute diameter) (not for NASA use) shall be 1,000 ohms. Resistors below 96.3 ohms in value shall be wound with .00175 inch (0.0445 mm) or larger diameter wire.



Standard style	Dimensions			
	A ±.031	B,min	C ±.0015	D ±.020
RWS81	.250	1.500	.0200	.085

Inches	mm
.0015	0.038
.0200	0.508
.031	1.79
.062	1.57
.085	2.16
.250	6.35
1.500	38.10

NOTES:
1. Dimensions are inches.
2. Metric equivalents are given for general information only.
3. Dimension "A" is "clean lead to clean lead".
4. Lead length for tape and reel packaging shall be 1 inch minimum.

FIGURE 1. Style RWS81 resistor.

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3.4.1 Maximum resistance value for "N" and "Z" type resistors. For resistors having noninductive characteristic "N" and "Z" designations, the maximum resistance value shall be 234 ohms for .001 inch (0.03 mm) nominal diameter wire. For .0008 inch (0.020 mm) nominal diameter wire (not for NASA use), the maximum resistance value shall be 499 ohms.

3.5 Power rating. The power rating shall be 1 watt at 25°C.

3.6 Weight. The maximum weight shall be .00077 pound (.35 gram); for "N" type resistors, the maximum weight shall be .00154 pound (.70 gram).

3.7 Terminal lead types. This style resistor is available in both solderable (S and N) and weldable (W and Z) leads.

3.8 Marking. Due to size limitation, this style resistor shall be marked with the minimum information in the order shown:

7333 - Date code
49R9 - Resistance
SFMJ - *Terminal designator, tolerance, failure rate, JAN marking*

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-83579.

4.2 Short - time overload. The duration of applied voltage shall be 5 seconds. Voltage applied shall be equivalent to 5 times rated wattage.

4.3 Dielectric withstanding voltage.

4.3.1 Atmospheric pressure. The magnitude of test voltage shall be 500 volts rms.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-83579.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-83579 are applicable to this specification.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 19
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 17, 85
DLA - ES

Preparing activity:
Air Force - 19

Agent:
DLA - ES

(Project 5905-1258-01)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-83579/1	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE), HIGH RELIABILITY, STYLE RWS81

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME SMC/SDFP P.O.C ALAN WANG	b. TELEPHONE (Include Area Code) (1) Commercial: 213-643-1966 (2) AUTOVON: 833-1966
c. ADDRESS (Include Zip Code) P.O. Box 92960 Los Angeles AFS Los Angeles, CA 9009-2960	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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